

# Simulation-based verification of **bfpr**

Samuel Pawel  
František Bartoš

Package version 0.3  
October 1, 2026

This report compares **bfpr** predictions with simulations across the listed priors, evidence thresholds, and sampling schedules.

## 1 $z$ -Test

### Simulated Design Priors

Table 1: Simulated design priors for the  $z$ -Test verification run. The Design column gives short IDs used by the analysis-prior table.

| Design | Design Prior     | Sequence          | Repetitions | USD |
|--------|------------------|-------------------|-------------|-----|
| Z1     | Point(0)         | 10 to 500 by 10   | 10000       | 1   |
| Z2     | Point(0.2)       | 10 to 500 by 10   | 10000       | 1   |
| Z3     | Point(0.5)       | 10 to 500 by 10   | 10000       | 1   |
| Z4     | Normal(0.5, 0.1) | 10 to 500 by 10   | 10000       | 1   |
| Z5     | Normal(0.2, 0.3) | 10 to 500 by 10   | 10000       | 1   |
| Z6     | Normal(0.2, 0.5) | 10 to 500 by 10   | 10000       | 1   |
| Z7     | Normal(0, 0.5)   | 10 to 500 by 10   | 10000       | 1   |
| Z8     | Point(-0.3)      | 10 to 500 by 10   | 10000       | 1   |
| Z9     | Point(0)         | 10 to 500 by 10   | 10000       | 50  |
| Z10    | Point(10)        | 10 to 500 by 10   | 10000       | 50  |
| Z11    | Normal(10, 20)   | 10 to 500 by 10   | 10000       | 50  |
| Z12    | Point(0)         | 10 to 10000 by 10 | 10000       | 1   |
| Z13    | Point(0.2)       | 10 to 10000 by 10 | 10000       | 1   |
| Z14    | Normal(0.2, 0.1) | 10 to 10000 by 10 | 10000       | 1   |
| Z15    | Normal(0.2, 0.5) | 10 to 10000 by 10 | 10000       | 1   |

### Analysis Priors

Table 2: Analysis priors used for the  $z$ -Test verification run. Designs lists the matching short IDs from the design-prior table.

| Function | H1                              | H0       | Designs               |
|----------|---------------------------------|----------|-----------------------|
| bf01     | Normal <sub>+</sub> (-0.2, 0.3) | Point(0) | Z1-Z5, Z7-Z8          |
| bf01     | Normal <sub>+</sub> (0, 0.1)    | Point(0) | Z1-Z5, Z7-Z8          |
| bf01     | Normal <sub>+</sub> (0, 0.707)  | Point(0) | Z1-Z5, Z7-Z8, Z12-Z13 |
| bf01     | Normal <sub>+</sub> (0, 2)      | Point(0) | Z1-Z5, Z7-Z8          |
| bf01     | Normal <sub>+</sub> (0.2, 0.3)  | Point(0) | Z1-Z5, Z7-Z8          |

|         |                                         |                                         |                       |
|---------|-----------------------------------------|-----------------------------------------|-----------------------|
| bf01    | Normal <sub>-</sub> (-0.2, 0.3)         | Point(0)                                | Z1-Z5, Z7-Z8          |
| bf01    | Normal <sub>-</sub> (0, 0.1)            | Point(0)                                | Z1-Z5, Z7-Z8          |
| bf01    | Normal <sub>-</sub> (0, 0.707)          | Point(0)                                | Z1-Z5, Z7-Z8, Z12-Z13 |
| bf01    | Normal <sub>-</sub> (0, 2)              | Point(0)                                | Z1-Z5, Z7-Z8          |
| bf01    | Normal <sub>-</sub> (0.2, 0.3)          | Point(0)                                | Z1-Z5, Z7-Z8          |
| bf01    | Normal(0, 0.707)                        | Point(0)                                | Z1-Z8, Z12-Z15        |
| bf01    | Normal(0, 20)                           | Point(0)                                | Z9-Z11                |
| bf01    | Normal(0.2, 0.3)                        | Point(0)                                | Z1-Z8, Z12-Z15        |
| bf01    | Normal(0.2, 0.707)                      | Point(0.2)                              | Z2-Z6, Z13-Z15        |
| bf01    | Normal(10, 20)                          | Point(0)                                | Z9-Z11                |
| bf01    | Point(0.2)                              | Point(0)                                | Z1-Z8, Z12-Z15        |
| bf01    | Point(0.5)                              | Point(0)                                | Z1-Z8, Z12-Z15        |
| bf01    | Point(0.5)                              | Point(0.2)                              | Z2-Z6, Z13-Z15        |
| bf01    | Point(10)                               | Point(0)                                | Z9-Z11                |
| dirbf01 | Normal <sub>+</sub> (0, 0.707)          | Normal <sub>-</sub> (0, 0.707)          | Z1-Z8, Z12-Z15        |
| dirbf01 | Normal <sub>+</sub> (0, 20)             | Normal <sub>-</sub> (0, 20)             | Z9-Z11                |
| dirbf01 | Normal <sub>+</sub> (0.2, 0.707; > 0.2) | Normal <sub>-</sub> (0.2, 0.707; ≤ 0.2) | Z2-Z6, Z13-Z15        |
| nmbf01  | Moment(0.354)                           | Point(0)                                | Z1-Z8, Z12-Z15        |
| nmbf01  | Moment(0.354)                           | Point(0.2)                              | Z2-Z6, Z13-Z15        |
| nmbf01  | Moment(0.707)                           | Point(0)                                | Z1-Z8, Z12-Z15        |
| nmbf01  | Moment(7.071)                           | Point(0)                                | Z9-Z11                |

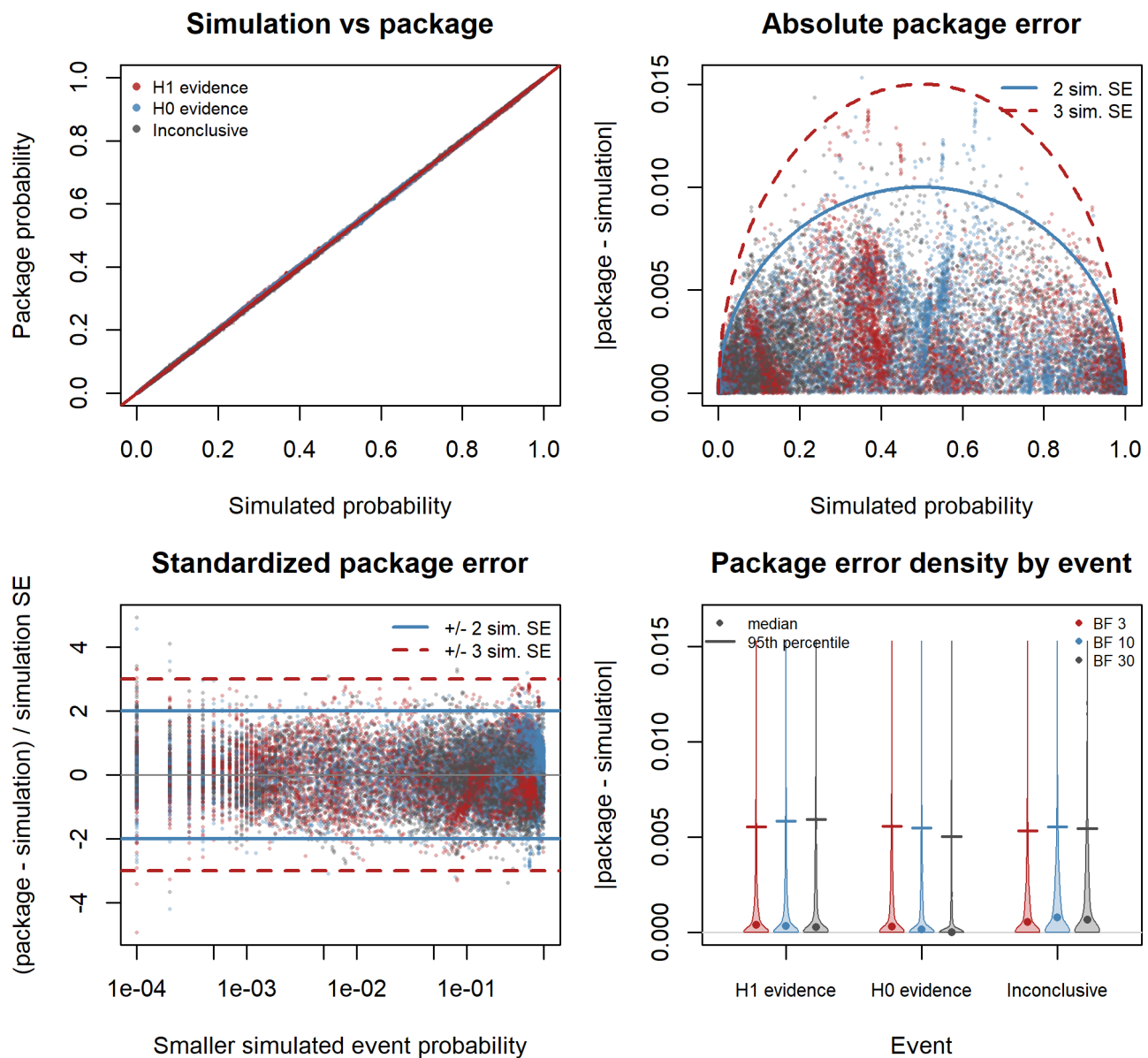
Normal( $\mu$ ,  $\sigma$ ) uses mean  $\mu$  and SD  $\sigma$  before truncation. Subscripts + and - denote truncation above and below zero; a nonzero cutoff is shown explicitly. Normal design priors remain untruncated.

## 1.1 Fixed-Design

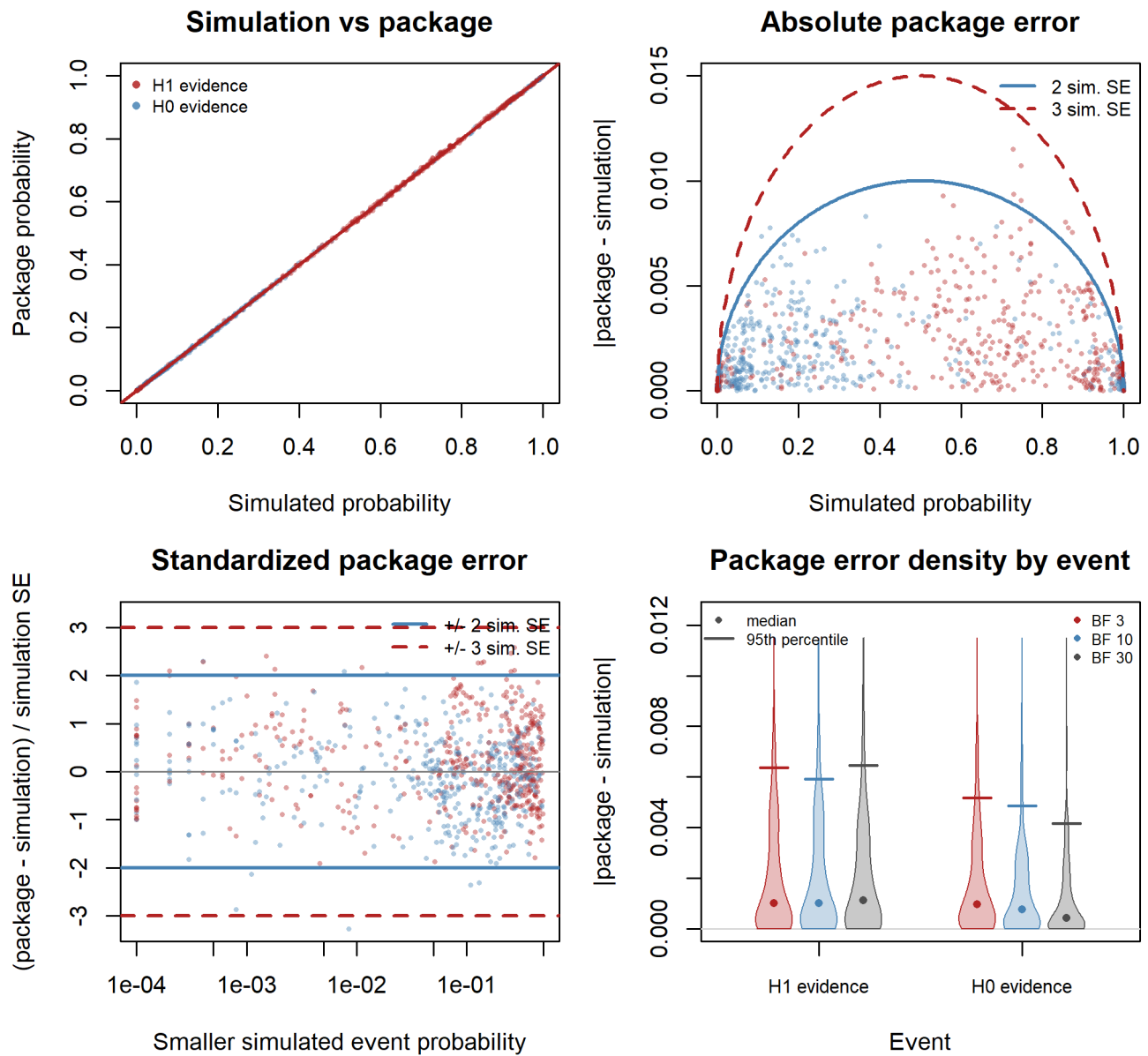
Table 3: Fixed-design verification settings for z-Test analyses. Designs refers to the design-prior table.

| Function                                          | Prior/design pairs | Designs      | Evidence thresholds | Sample sizes                 |
|---------------------------------------------------|--------------------|--------------|---------------------|------------------------------|
| bf01                                              | 51                 | Z1-Z11       | BF 3, 10, 30        | 30, 130, 260, 380, 500       |
| bf01                                              | 22                 | Z12-Z15      | BF 3, 10, 30        | 510, 2510, 5000, 7500, 10000 |
| nmbf01                                            | 24                 | Z1-Z11       | BF 3, 10, 30        | 30, 130, 260, 380, 500       |
| nmbf01                                            | 11                 | Z12-Z15      | BF 3, 10, 30        | 510, 2510, 5000, 7500, 10000 |
| dirbf01                                           | 16                 | Z1-Z11       | BF 3, 10, 30        | 30, 130, 260, 380, 500       |
| dirbf01                                           | 7                  | Z12-Z15      | BF 3, 10, 30        | 510, 2510, 5000, 7500, 10000 |
| bf01 (Normal <sub>+</sub> , Normal <sub>-</sub> ) | 70                 | Z1-Z5, Z7-Z8 | BF 3, 10, 30        | 10 to 500 by 10              |

### bf01 fixed-design checks



# nmbf01 fixed-design checks



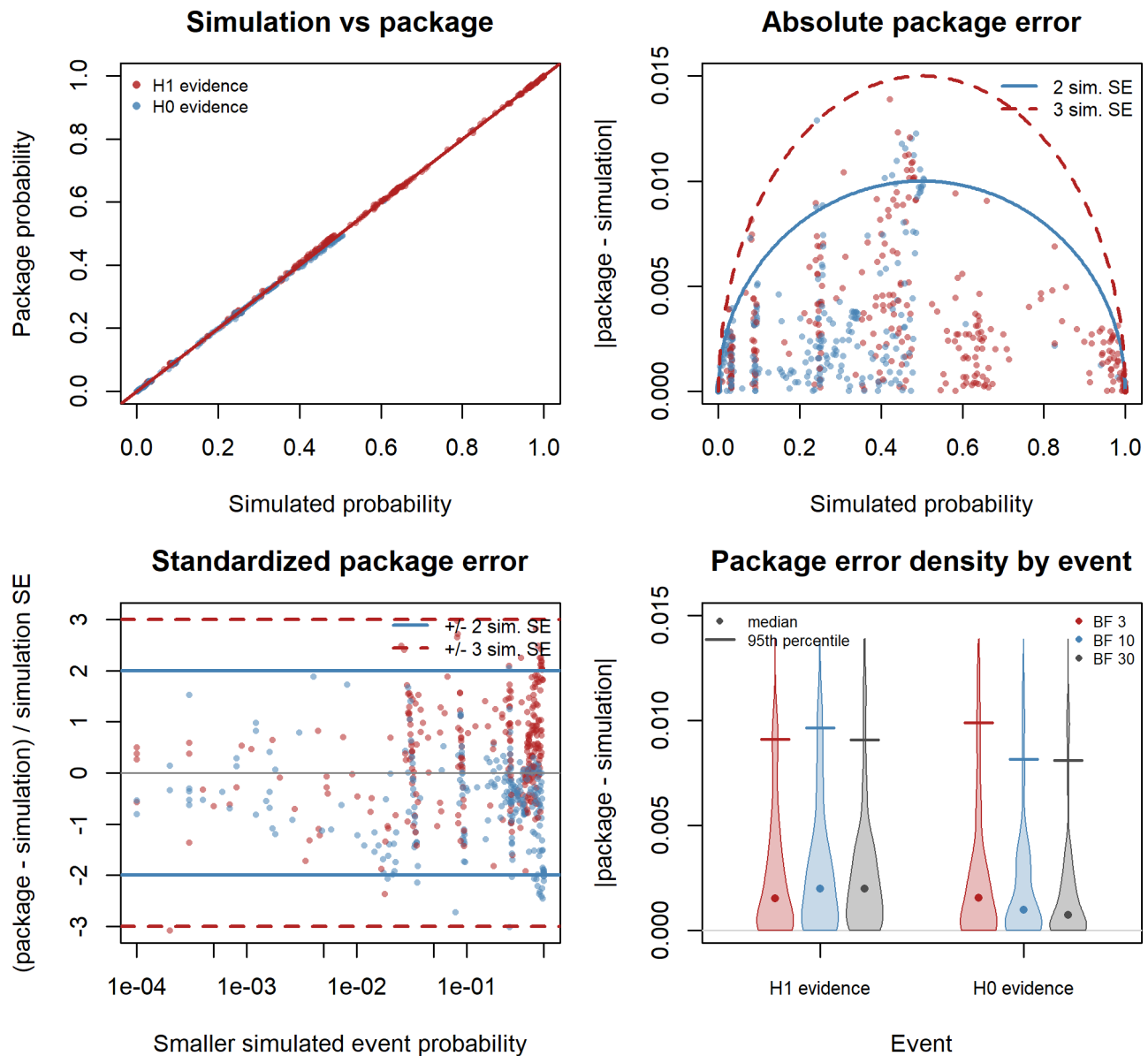
**dirbf01 fixed-design checks****1.2 Sequential-Design**

Table 4: Sequential-design verification settings for z-Test analyses. Designs refers to the design-prior table.

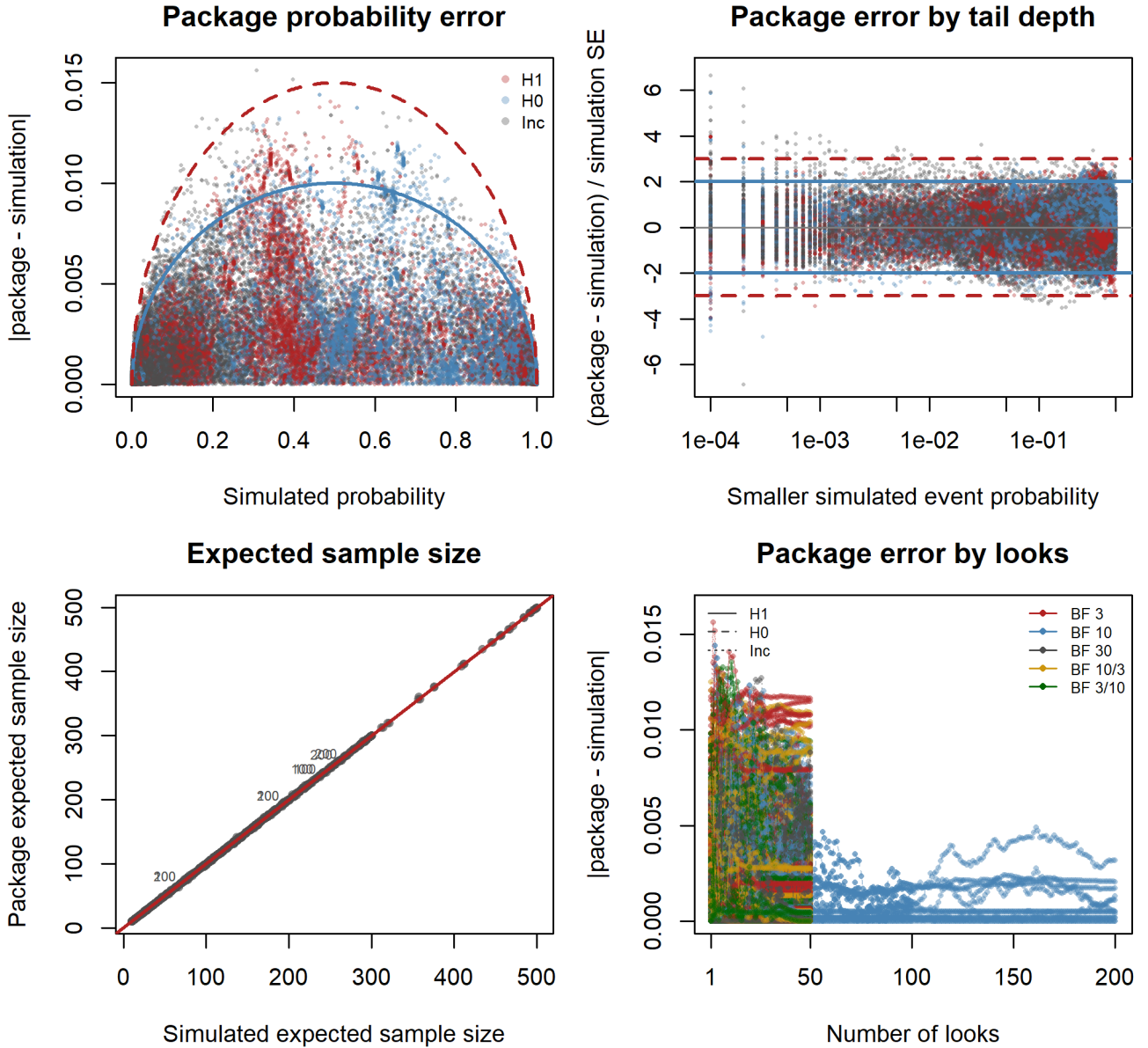
| Function                                          | Prior/design pairs | Designs                    | Evidence thresholds      | Look sequence                           |
|---------------------------------------------------|--------------------|----------------------------|--------------------------|-----------------------------------------|
| bf01                                              | 16                 | Z4-Z5, Z7, Z11-Z12, Z14    | BF 3, 10, 30             | start 20, by 10 (2, 5, 10 looks)        |
| nmbf01                                            | 16                 | Z4-Z5, Z7-Z8, Z11-Z12, Z14 | BF 3, 10, 30             | start 20, by 10 (2, 5, 10 looks)        |
| dirbf01                                           | 11                 | Z4-Z5, Z7-Z9, Z11-Z12, Z14 | BF 3, 10, 30             | start 20, by 10 (2, 5, 10 looks)        |
| bf01 (Normal <sub>+</sub> , Normal <sub>-</sub> ) | 74                 | Z1-Z5, Z7-Z8, Z12-Z13      | BF 3, 10, 30; 10/3, 3/10 | 1, 3, 50 looks; selected 100, 200 looks |

The Normal<sub>+</sub> and Normal<sub>-</sub> cases use a single look at 300, three equally spaced looks (100, 200, 300), three uneven looks (20, 100, 300), and 50 looks from 10 to 500. For the centered prior with SD 0.707 and point design priors at 0 and 0.2, schedules also contain 100 or 200 looks in steps of 10. BF 10/3 and BF 3/10 denote asymmetric stopping thresholds  $(k_1, k_0) = (1/10, 3)$  and  $(1/3, 10)$ .

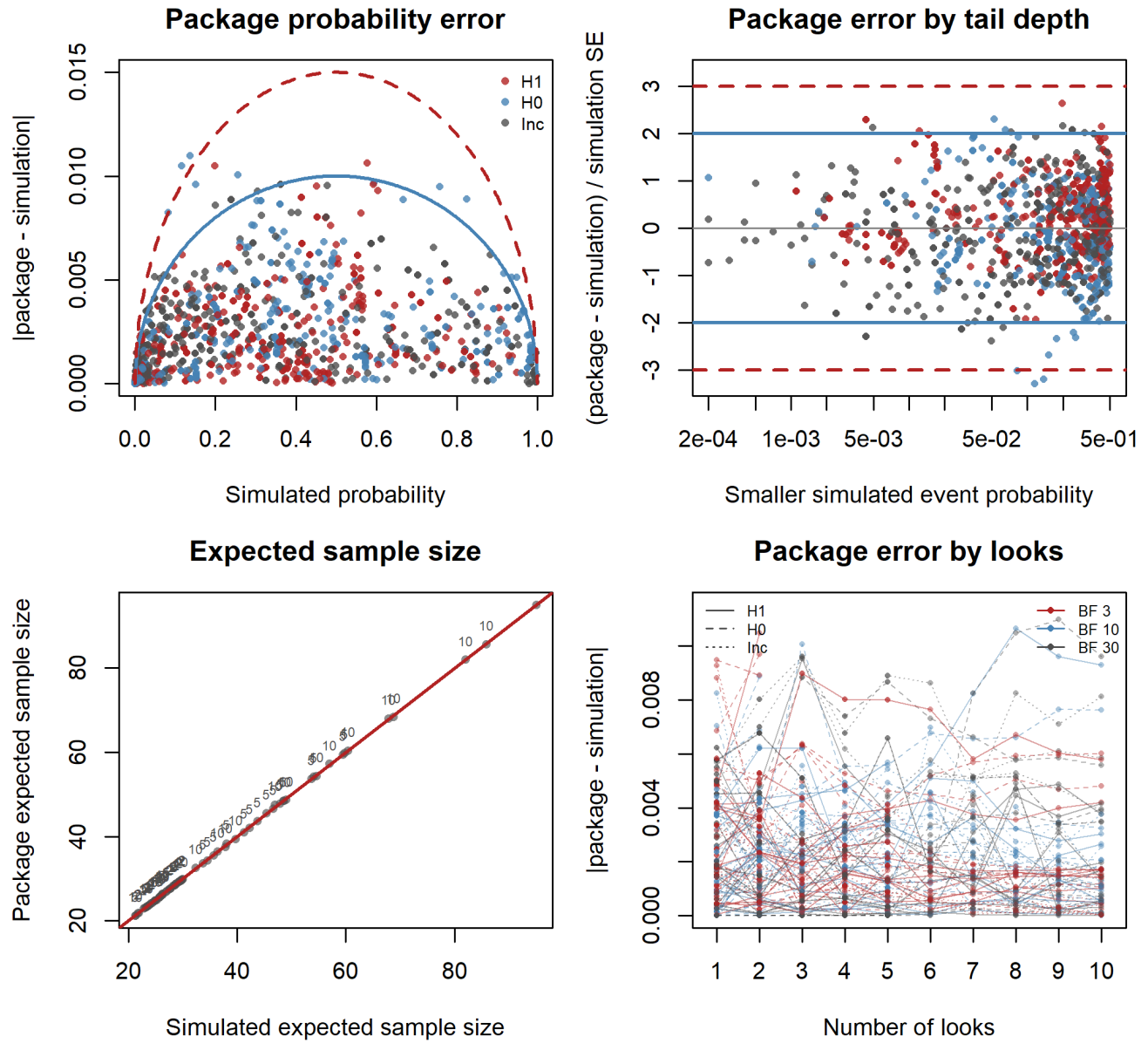
Very rare stopping probabilities can be missed by the integration grid. Events whose fixed-look union bound implies fewer than one expected occurrence in 10,000 trajectories are retained as accuracy diagnostics, including their original interval failures. Their relative accuracy is not certified; larger discrepancies and invalid probabilities remain required failures.

For example, the Normal<sub>-</sub>(0,0.1) analysis prior and Normal(0,0.5) design prior with  $(k_1, k_0) = (1/30, 30)$  give an H0 stopping prediction of  $2.22 \times 10^{-19}$  by  $n = 110$ , whereas the fixed-look union bound is  $1.18 \times 10^{-5}$ . The simulation contains 1 such event in 10,000 trajectories.

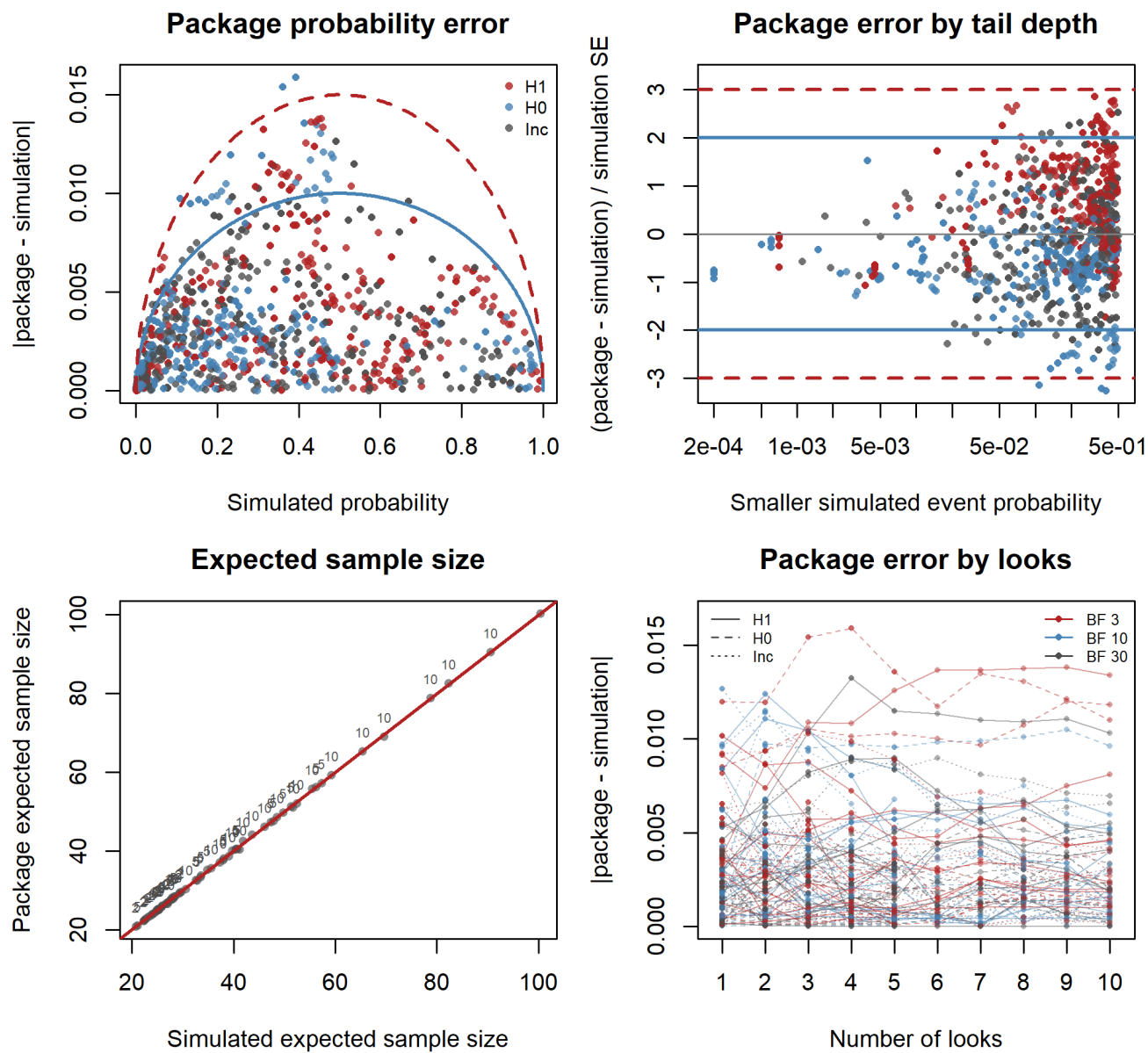
### bf01 sequential-design checks



### nmbf01 sequential-design checks



### dirbf01 sequential-design checks



## 2 *t*-Test

### Simulated Design Priors

Table 5: Simulated design priors for the *t*-Test verification run. The Design column gives short IDs used by the analysis-prior table.

| Design | Design Prior     | Sequence          | Repetitions | Sampling   | n2/n1 |
|--------|------------------|-------------------|-------------|------------|-------|
| T1     | Point(0)         | 10 to 500 by 10   | 10000       | two-sample | 1     |
| T2     | Point(0.5)       | 10 to 500 by 10   | 10000       | two-sample | 1     |
| T3     | Normal(0.5, 0.1) | 10 to 500 by 10   | 10000       | two-sample | 1     |
| T4     | Normal(0.5, 0.5) | 10 to 500 by 10   | 10000       | two-sample | 1     |
| T5     | Point(0)         | 10 to 500 by 10   | 10000       | one-sample |       |
| T6     | Point(0.5)       | 10 to 500 by 10   | 10000       | one-sample |       |
| T7     | Point(0)         | 10 to 500 by 10   | 10000       | paired     |       |
| T8     | Point(0.5)       | 10 to 500 by 10   | 10000       | paired     |       |
| T9     | Point(-0.4)      | 10 to 500 by 10   | 10000       | two-sample | 1     |
| T10    | Point(0)         | 10 to 500 by 10   | 10000       | two-sample | 1.1   |
| T11    | Point(0.35)      | 10 to 500 by 10   | 10000       | two-sample | 1.1   |
| T12    | Normal(0, 0.5)   | 10 to 500 by 10   | 10000       | two-sample | 1     |
| T13    | Point(0)         | 10 to 10000 by 10 | 10000       | two-sample | 1     |
| T14    | Point(0.2)       | 10 to 10000 by 10 | 10000       | two-sample | 1     |
| T15    | Normal(0.2, 0.1) | 10 to 10000 by 10 | 10000       | two-sample | 1     |
| T16    | Normal(0.2, 0.5) | 10 to 10000 by 10 | 10000       | two-sample | 1     |

### Analysis Priors

Table 6: Analysis priors used for the *t*-Test verification run. Designs lists the matching short IDs from the design-prior table.

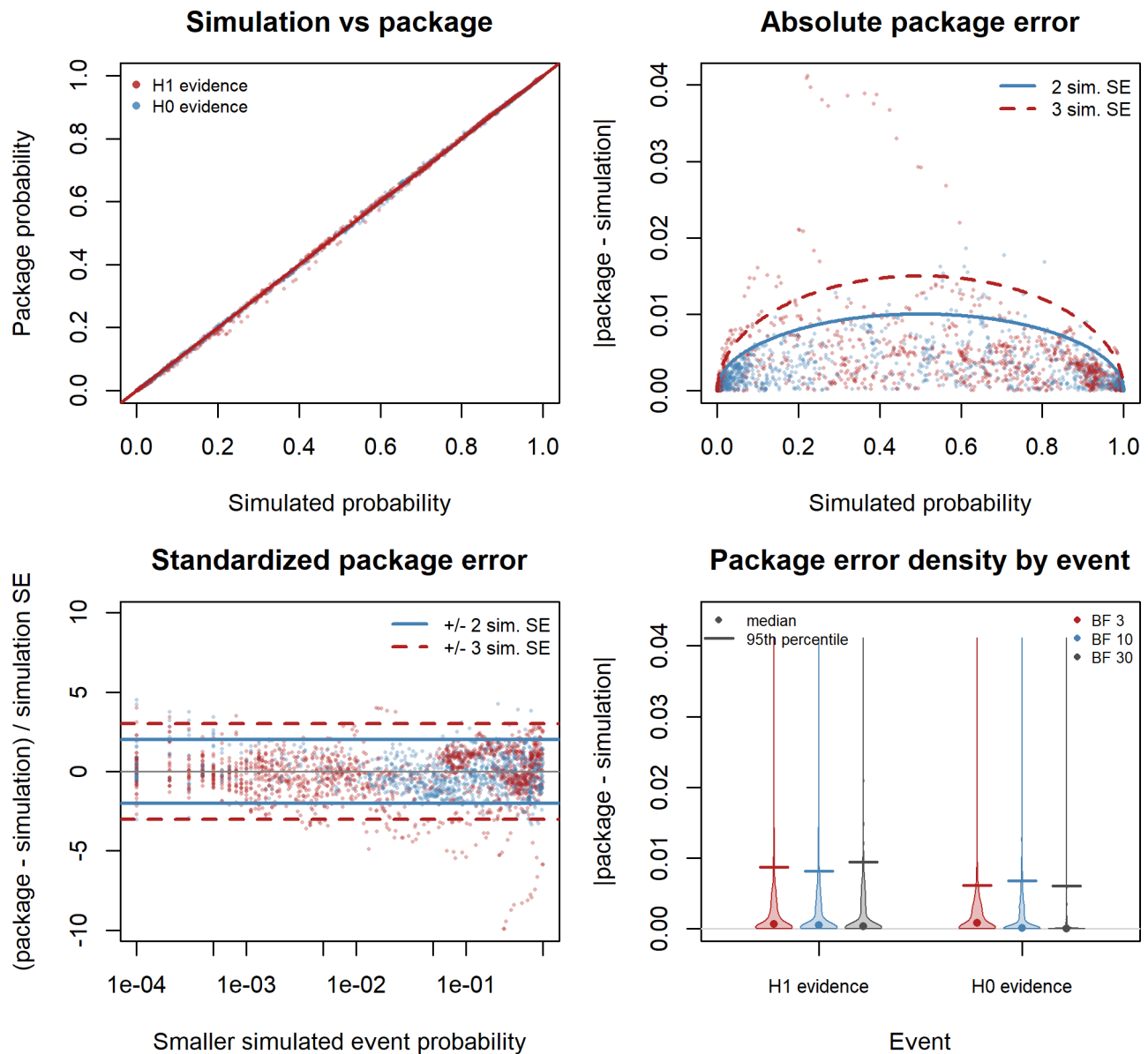
| Function | H1                                           | H0         | Designs                     |
|----------|----------------------------------------------|------------|-----------------------------|
| tbfbf01  | Cauchy <sub>+</sub> (0, 0.707)               | Point(0)   | T1-T16                      |
| tbfbf01  | Cauchy <sub>-</sub> (0, 0.707)               | Point(0)   | T1-T16                      |
| tbfbf01  | Cauchy(0, 0.35)                              | Point(0)   | T1-T16                      |
| tbfbf01  | Cauchy(0, 0.707)                             | Point(0)   | T1-T16                      |
| tbfbf01  | Cauchy(0, 1)                                 | Point(0)   | T1-T16                      |
| tbfbf01  | StudentT <sub>+</sub> (3, 0.5, 0.1)          | Point(0)   | T2-T4, T6, T8, T11, T14-T16 |
| tbfbf01  | StudentT <sub>+</sub> (3, 0.5, 0.35; > 0.2)  | Point(0.2) | T11, T14-T16                |
| tbfbf01  | StudentT <sub>+</sub> (30, 0.5, 0.35; > 0.2) | Point(0.2) | T1, T5, T7, T10, T13        |
| tbfbf01  | StudentT(3, 0, 0.707)                        | Point(0)   | T1-T16                      |
| tbfbf01  | StudentT(30, 0, 0.707)                       | Point(0)   | T1-T16                      |

## 2.1 Fixed-Design

Table 7: Fixed-design verification settings for t-Test analyses. Designs refers to the design-prior table.

| Function | Prior/design pairs | Designs | Evidence thresholds | Sample sizes                 |
|----------|--------------------|---------|---------------------|------------------------------|
| tbf01    | 95                 | T1-T12  | BF 3, 10, 30        | 30, 130, 260, 380, 500       |
| tbf01    | 35                 | T13-T16 | BF 3, 10, 30        | 510, 2510, 5000, 7500, 10000 |

### tbf01 fixed-design checks



## 2.2 Sequential-Design

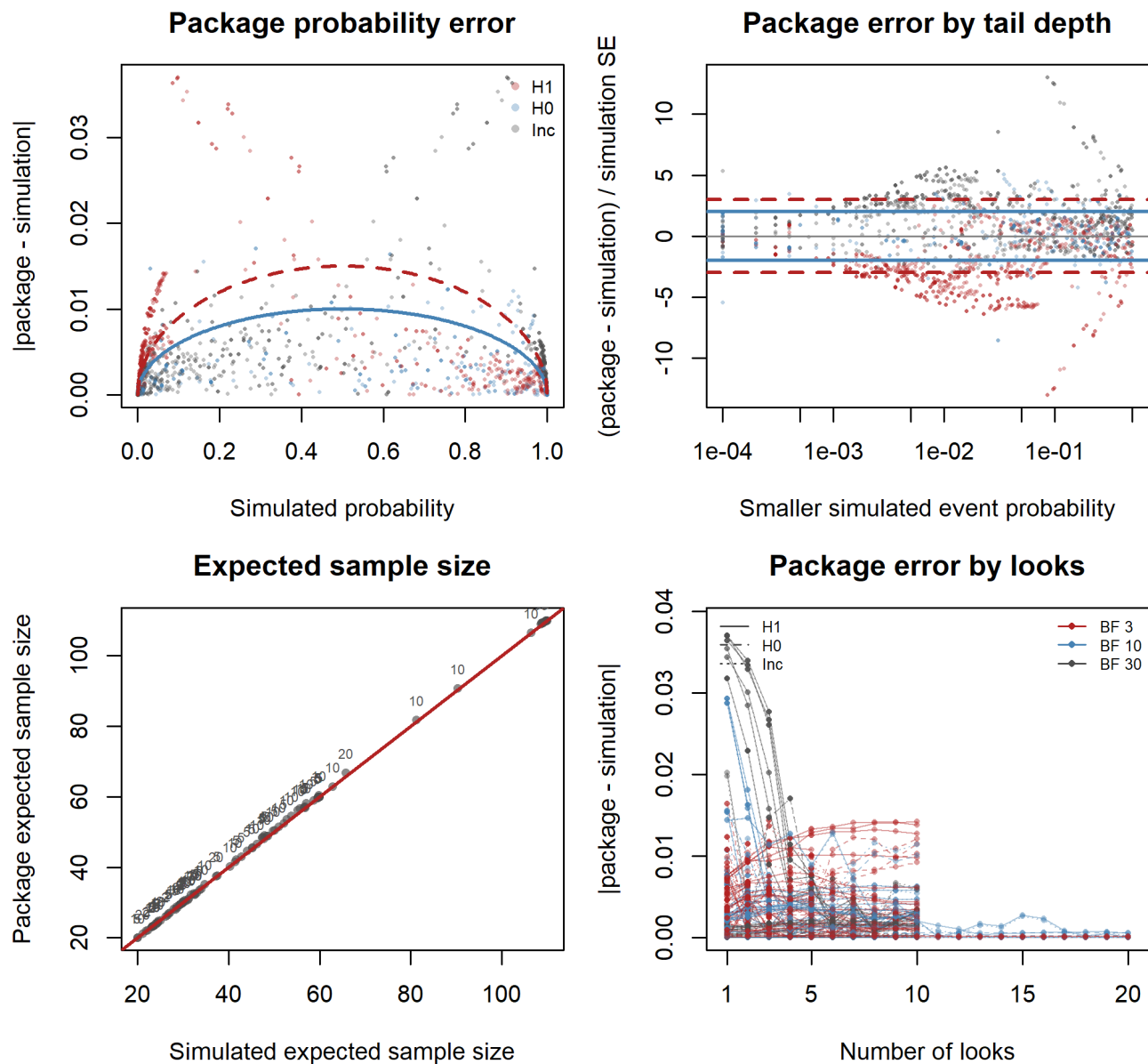
Table 8: Sequential-design verification settings for t-Test analyses. Designs refers to the design-prior table.

| Function | Prior/design pairs | Designs            | Evidence thresholds | Look sequence                        |
|----------|--------------------|--------------------|---------------------|--------------------------------------|
| tbf01    | 27                 | T2, T5-T6, T9, T13 | BF 3, 10, 30        | start 20, by 10 (2, 5, 10, 20 looks) |

The sequential *t*-test probabilities use a normal approximation to the sampling distribution of the test statistics. Comparisons with exact-*t* simulations therefore diagnose this approximation; they do not certify exact-*t* accuracy. Numerical validity and fixture integrity remain required checks.

In this run, 199 of 2688 probability comparisons fall outside the severe binomial prediction intervals; the largest absolute difference is 3.703 percentage points.

### tbf01 sequential-design checks



### 3 Binomial Test

#### Simulated Design Priors

Table 9: Simulated design priors for the Binomial Test verification run. The Design column gives short IDs used by the analysis-prior table.

| Design | Design Prior                    | Sequence          | Repetitions |
|--------|---------------------------------|-------------------|-------------|
| B1     | Point(0.5)                      | 10 to 500 by 10   | 10000       |
| B2     | Point(0.6)                      | 10 to 500 by 10   | 10000       |
| B3     | Point(0.2)                      | 10 to 500 by 10   | 10000       |
| B4     | Point(0.75)                     | 10 to 500 by 10   | 10000       |
| B5     | Beta(60, 40)                    | 10 to 500 by 10   | 10000       |
| B6     | Beta <sub>+</sub> (1, 1; > 0.5) | 10 to 500 by 10   | 10000       |
| B7     | Beta <sub>-</sub> (1, 1; ≤ 0.5) | 10 to 500 by 10   | 10000       |
| B8     | Point(0.5)                      | 10 to 10000 by 10 | 10000       |
| B9     | Point(0.55)                     | 10 to 10000 by 10 | 10000       |

#### Analysis Priors

Table 10: Analysis priors used for the Binomial Test verification run. Designs lists the matching short IDs from the design-prior table.

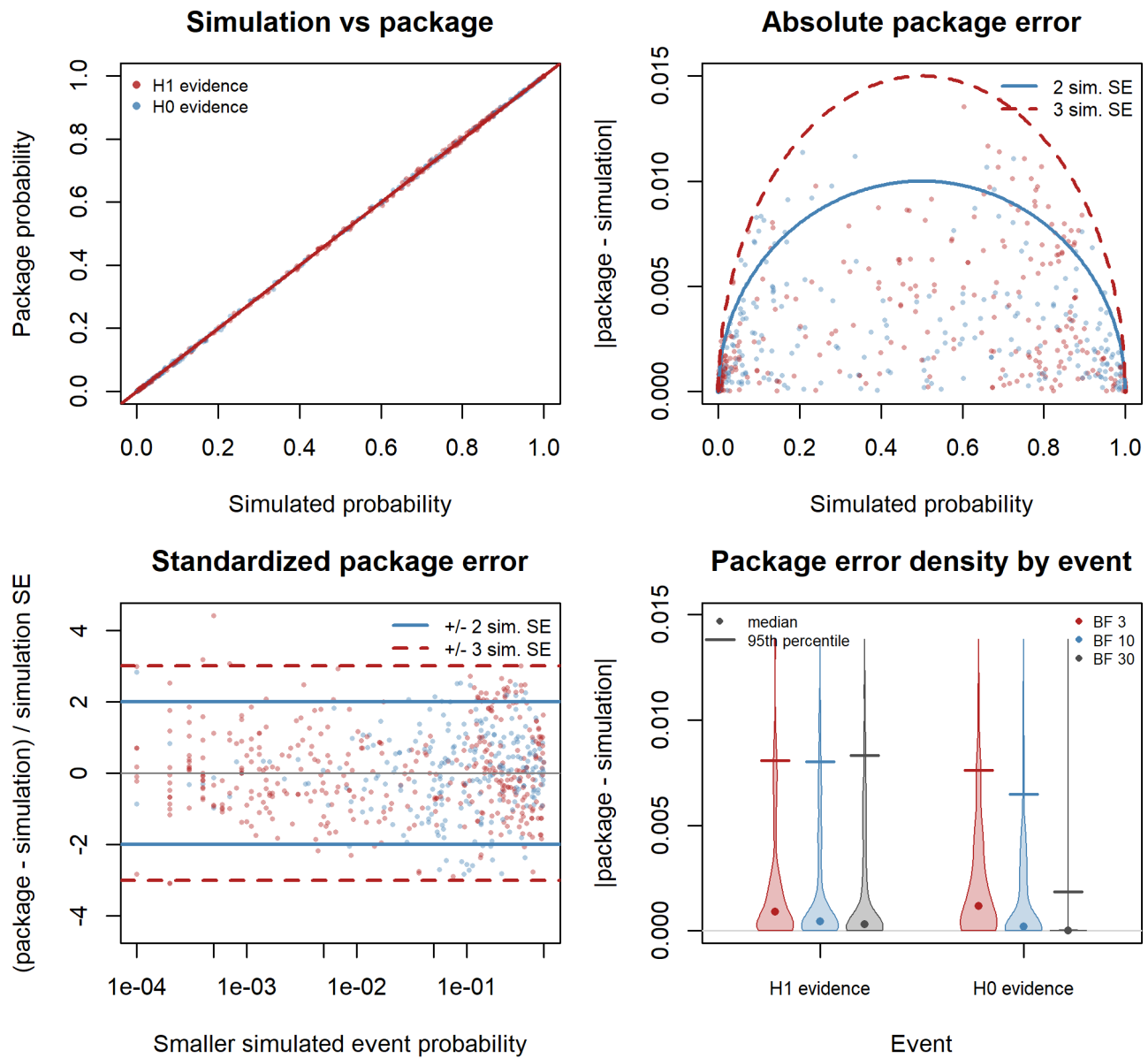
| Function            | H1                               | H0                               | Designs |
|---------------------|----------------------------------|----------------------------------|---------|
| binbf01 (direction) | Beta <sub>+</sub> (1, 1; > 0.2)  | Beta <sub>-</sub> (1, 1; ≤ 0.2)  | B3      |
| binbf01 (direction) | Beta <sub>+</sub> (1, 1; > 0.5)  | Beta <sub>-</sub> (1, 1; ≤ 0.5)  | B1-B9   |
| binbf01 (direction) | Beta <sub>+</sub> (1, 1; > 0.55) | Beta <sub>-</sub> (1, 1; ≤ 0.55) | B9      |
| binbf01 (direction) | Beta <sub>+</sub> (1, 1; > 0.6)  | Beta <sub>-</sub> (1, 1; ≤ 0.6)  | B2, B5  |
| binbf01 (direction) | Beta <sub>+</sub> (1, 1; > 0.75) | Beta <sub>-</sub> (1, 1; ≤ 0.75) | B4      |
| binbf01 (point)     | Beta(1, 1)                       | Point(0.2)                       | B3      |
| binbf01 (point)     | Beta(1, 1)                       | Point(0.5)                       | B1-B9   |
| binbf01 (point)     | Beta(1, 1)                       | Point(0.55)                      | B9      |
| binbf01 (point)     | Beta(1, 1)                       | Point(0.6)                       | B2, B5  |
| binbf01 (point)     | Beta(1, 1)                       | Point(0.75)                      | B4      |
| binbf01 (point)     | Beta(40, 60)                     | Point(0.5)                       | B1-B9   |
| binbf01 (point)     | Beta(5100, 4900)                 | Point(0.5)                       | B8-B9   |
| binbf01 (point)     | Beta(60, 40)                     | Point(0.5)                       | B1-B9   |

#### 3.1 Fixed-Design

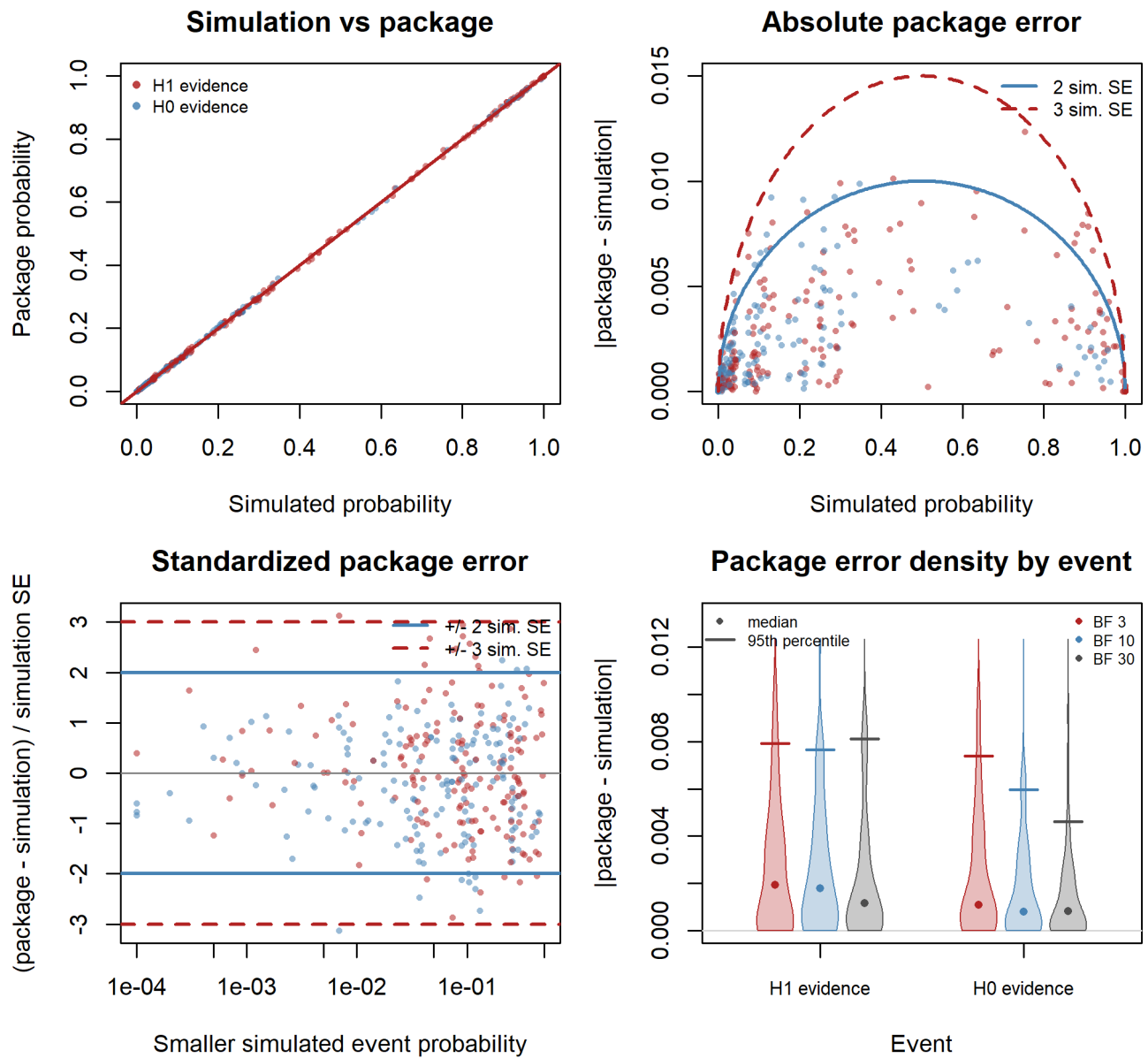
Table 11: Fixed-design verification settings for Binomial Test analyses. Designs refers to the design-prior table.

| Function            | Prior/design pairs | Designs | Evidence thresholds | Sample sizes                 |
|---------------------|--------------------|---------|---------------------|------------------------------|
| binbf01 (point)     | 25                 | B1-B7   | BF 3, 10, 30        | 30, 130, 260, 380, 500       |
| binbf01 (point)     | 9                  | B8-B9   | BF 3, 10, 30        | 510, 2510, 5000, 7500, 10000 |
| binbf01 (direction) | 11                 | B1-B7   | BF 3, 10, 30        | 30, 130, 260, 380, 500       |
| binbf01 (direction) | 3                  | B8-B9   | BF 3, 10, 30        | 510, 2510, 5000, 7500, 10000 |

### binbf01-point fixed-design checks



### binbf01-direction fixed-design checks



## 4 Sample-Size Search Checks

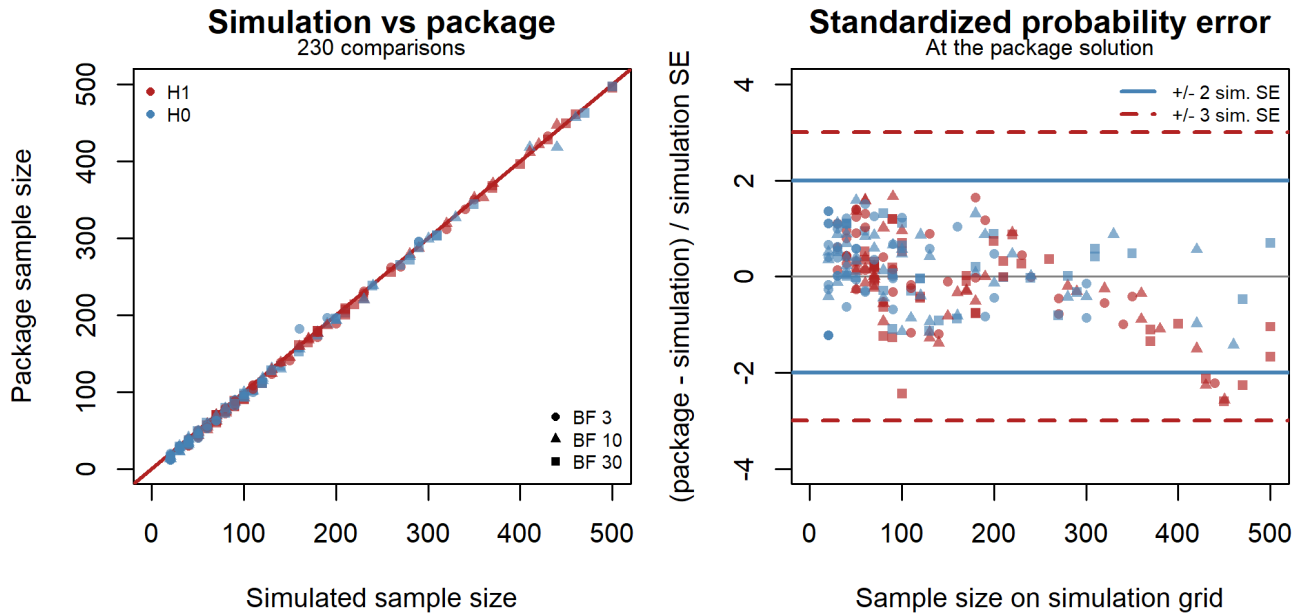
The searches cover `nbf01` with one-sided Normal priors, `nbf01seq`, and `ntbf01seq`. They compare the first sample sizes at which simulated and computed probabilities of evidence for H1 or H0 reach a target. Sequential sample sizes refer to the maximum planned sample size. Only finite results from both methods enter the figures.

Left panels compare sample sizes. Right panels compare computed and simulated probabilities at the package solution, expressed in Monte Carlo SEs, with reference lines at  $\pm 2$  and  $\pm 3$  SEs. For fixed designs, the package solution is rounded up to the simulation grid for the probability comparison.

### 4.1 Fixed-Design z-Test: `nbf01`

These comparisons cover the 70 Normal<sub>+</sub> and Normal<sub>-</sub> analysis-prior/design-prior combinations listed in the z-test section. Targets are 0.8 and 0.9 for both H1 and H0, with BF thresholds 3, 10, and 30 and a search range from 10 to 500. The package searches over integer sample sizes; simulated probabilities are available every 10 observations.

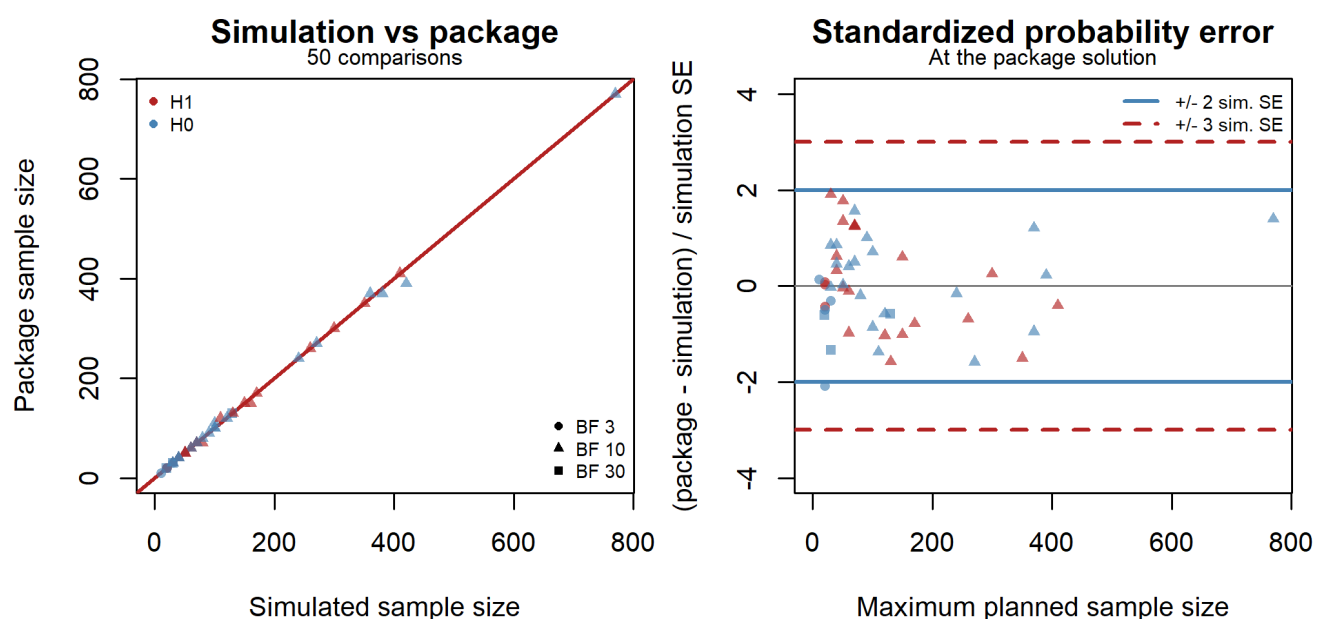
#### `nbf01`: one-sided normal



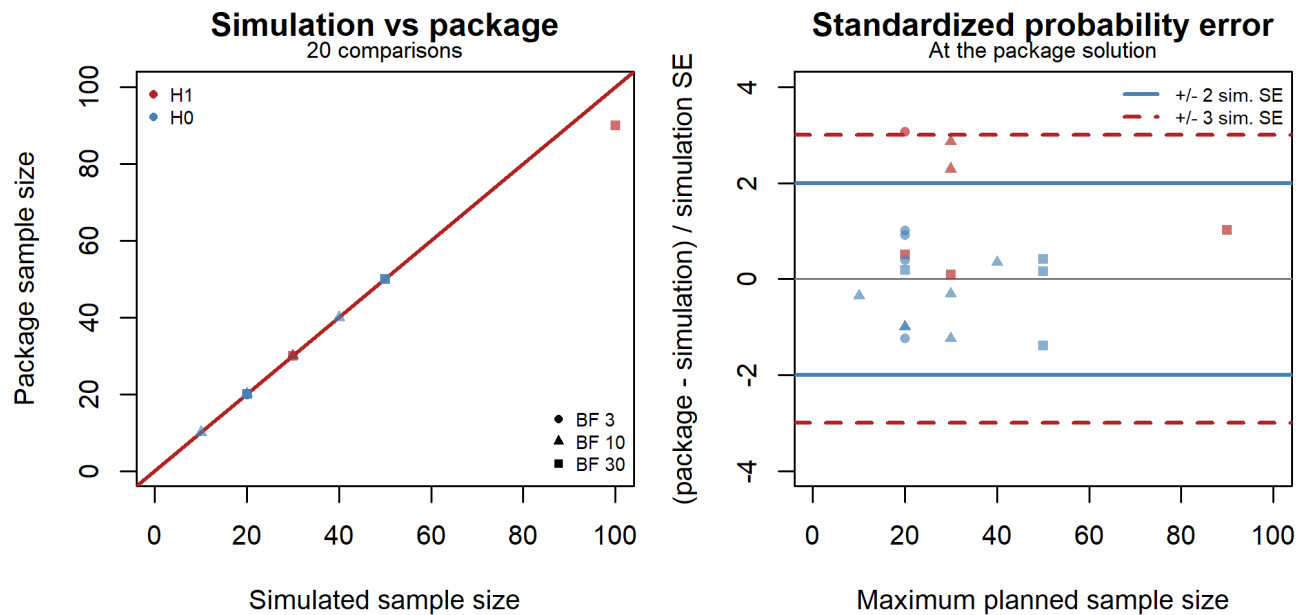
## 4.2 Sequential z-Test: nbf01seq

The normal and point priors, normal moment priors, and directional hypotheses use 15, 20, and 15 searches, respectively, with the analysis and design priors listed in the z-test section. Targets are 0.1, 0.3, and 0.8 for all three models, and 0.95 for directional hypotheses. BF thresholds are 3, 10, and 30 for H1 and H0. Looks occur every 10 observations starting at 10 or 20. Upper search bounds depend on the scenario and extend to 2,000 for normal and directional models and 1,000 for moment priors. The one-sided normal cases use  $\text{Normal}_+(0, 0.707)$ ,  $\text{Normal}_-(0, 0.707)$ ,  $\text{Normal}_+(0.2, 0.3)$ , and  $\text{Normal}_-(0.2, 0.3)$ , all seven generating distributions, and targets 0.8 and 0.9 for H1 and H0. These 112 searches use  $(k_1, k_0) = (1/10, 10)$  and looks from 10 to 500 in steps of 10.

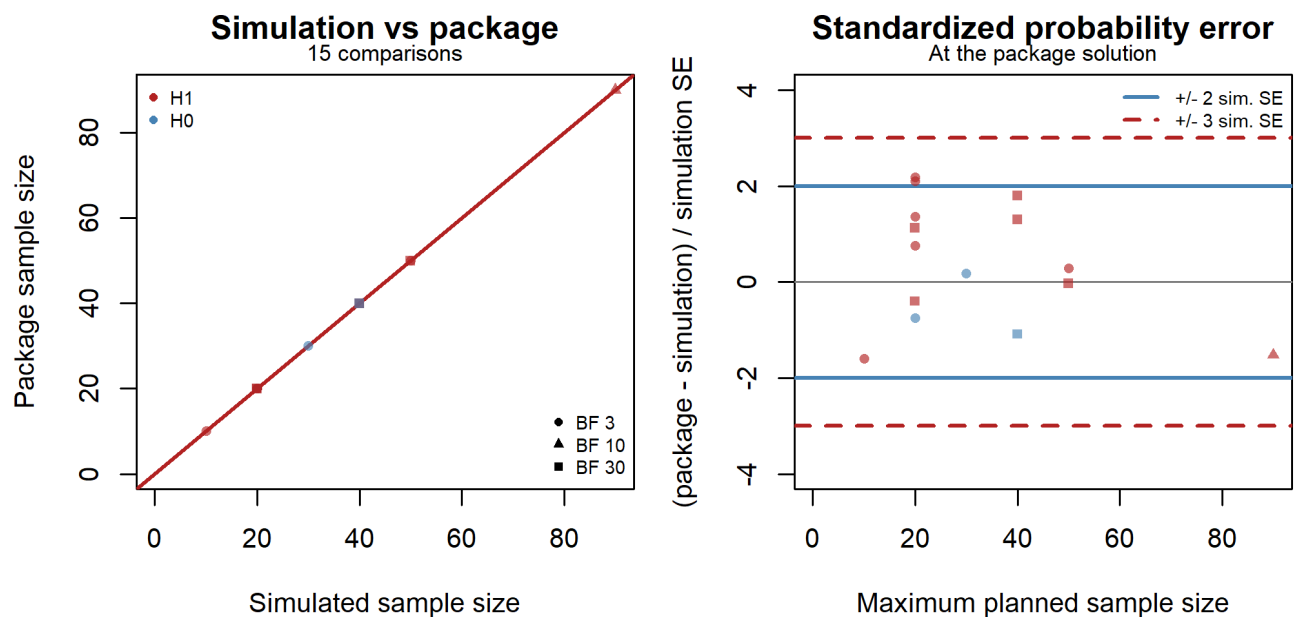
### nbf01seq: normal and point priors



### nbf01seq: normal moment prior



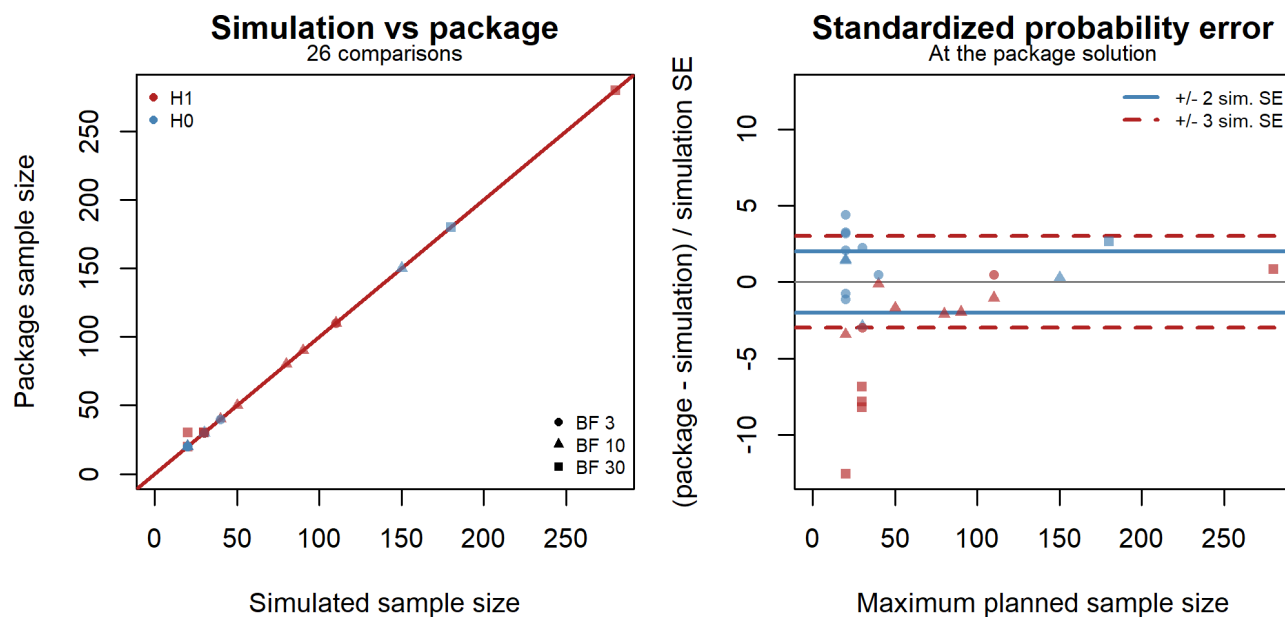
### nbf01seq: directional hypotheses



### 4.3 Sequential t-Test: *ntbf01seq*

The 26 searches use Cauchy and StudentT analysis priors with two-sided, greater, and less alternatives, and point and Normal design priors. Sampling is one-sample, paired, or two-sample, with two-sample allocation ratios 1 and 1.1. Targets are 0.1, 0.3, 0.8, and 0.95, with BF thresholds 3, 10, and 30 for H1 and H0. Looks occur every 10 observations starting at 10 or 20, and upper search bounds range from 30 to 2,000. Two-sample sizes refer to group 1. For allocation ratio 1.1, group-2 sizes are rounded to the nearest integer in the simulation and upward in the package search. The sequential t probabilities use a normal approximation to the sampling distribution of the test statistics.

#### *ntbf01seq*: t-test priors



## 5 Run Time Diagnostics

Elapsed times are recorded for the package calculations shown above. Fixed-design probability calls can cover several rows at once because those functions are vectorized; most sequential and sample-size search calls are one setting at a time.

Calculations used 6 concurrent workers.

Table 12: Probability and sample-size search run times. Fixed probability rows can be vectorized, so calls and rows covered need not match.

| Function  | Calls | Rows covered | Median  | P90     | Max       | Total      |
|-----------|-------|--------------|---------|---------|-----------|------------|
| pbf01     | 73    | 2,190        | 0 s     | 0.02 s  | 0.09 s    | 0.42 s     |
| pbf01seq  | 239   | 906          | 0.08 s  | 2.464 s | 1.347 min | 23.796 min |
| pbinbf01  | 48    | 1,440        | 0.06 s  | 0.09 s  | 0.33 s    | 3.36 s     |
| pnmbf01   | 35    | 1,050        | 0 s     | 0.02 s  | 0.36 s    | 0.47 s     |
| ptbf01    | 130   | 3,900        | 1.405 s | 2.624 s | 3.06 s    | 3.001 min  |
| ptbf01seq | 148   | 148          | 0.61 s  | 3.866 s | 1.592 min | 13.201 min |
| nbf01seq  | 50    | 50           | 0.015 s | 0.474 s | 1.584 min | 1.921 min  |
| ntbf01seq | 26    | 26           | 0.195 s | 3.55 s  | 11.91 s   | 31.12 s    |

Table 13: One-sided z-test run times. Each fixed pbf01 run times the H1 and H0 vector calls together; each other run times one package call.

| Function | Timed runs | Median  | P90      | Max       | Total      |
|----------|------------|---------|----------|-----------|------------|
| nbf01    | 840        | 0.01 s  | 0.02 s   | 0.13 s    | 8.05 s     |
| nbf01seq | 112        | 8.855 s | 10.777 s | 12.08 s   | 12.966 min |
| pbf01    | 210        | 0.03 s  | 0.04 s   | 0.11 s    | 6.03 s     |
| pbf01seq | 1408       | 0.03 s  | 10.09 s  | 2.993 min | 69.321 min |

Histograms show the calls in the probability and sample-size search timing table.

### Package Function Call Timing

